Nova Bank fictional bank

CASE STUDY

Nova Support Copilot

The right answer about your own fee, dispute or card, in your language, with a person one tap away. PRD

An AI support copilot for a fictional Egyptian bank. It answers from the customer's own records and the bank's rules, and hands fraud, complaints and anything unsure to a person. PRD STATUS

Project report (map) · PRD · Prototype (runs locally) · Demo Day slides · All documents · ROI model + simulator · 44 assumptions · Devil's advocate review · Delivery playbook · Code on GitHub

Sherif Hilal · AI Product Management capstone · November 2026 · version 1.0.15, built and tested on synthetic data, stopped before the internal beta STATUS

At a glance

45K	contacts a month, 120 agents PRD D1	60%	are the same 3 questions PRD D2
106	release-test cases: SHIP STATUS	13/13	fraud cases sent to Fraud STATUS
97%	answers grounded (33/34) STATUS	$2.46	total AI spend on testing STATUS
63%	shadow-run answer rate, not resolution STATUS	16	scored versions, 1.0.0 to 1.0.15 STATUS
5.5s	slowest 5% of replies; pilot limit 6 s STATUS	0	real users so far STATUS

Status: built and tested end to end on synthetic data. Everything left needs real people STATUS.

How to read the tags

D3 C1 R1 … are assumption IDs from the Stage 1 assumptions list. A5 … are challenges from the devil's advocate review. Document tags: PRD PRD · STATUS project status · ROI ROI model · MKT market and competition · GUIDE project guide · REG regulatory research · UX UX principles · PLAY delivery playbook.

sample marks illustrative content. synthetic data marks real screens and results on made-up data. Most links open for the author only until shared.

CONTEXT

What is the product, and why does it matter?

What is the product?

- **Customer chat** in Arabic, English and Franco-Arabic, with sources under every answer PRD UX
- **Agent inbox** with Fraud, Complaints and Agent queues PRD
- **PM control panel** with key results and the ROI model STATUS
- Read-only: never moves money, waives fees or decides disputes PRD

✅ fees ✅ disputes ✅ cards ❌ loans, advice PRD

How does it earn?

An internal tool: it saves agent time, so "pricing" means cost to run and payback GUIDE.

+18% ROI if hiring really falls ROI

-56% ROI on minutes saved only ROI

Base case, assumptions. AI is about EGP 81 a month; running and build cost EGP 93,333 B4 B5.

What industry is it in?

AI support chatbots for banks. Crowded globally, thin in Egypt. Competition score **7/10** MKT.

Growth factors: contacts grow 15% a year D4; Salesforce agreed to buy Intercom Fin (June 2026) MKT; Intella and Kore.ai move in the region MKT.

Banque Misr, NBE, CIB bots · Fin · Agentforce · Kore.ai · Intella · Erica · Klarna

Where Nova sits: type 3 of 6, grounded answers, with actions kept for people MKT.

Objective A logged-in customer gets a correct, sourced answer to the 3 in-scope questions, with no handoff and no repeat contact within 7 days. Target: **40% verified resolution.** PRD STATUS

The key moment: one month, step by step

- **45,000 contacts a month** D1
- **27,000 are the 3 questions (60%)** D2
- **5,400 start in chat (20%)** C1 ROI
- **2,160 resolved (40%)** C4 ROI

The moment that matters: a customer asks about their own money, in chat. Red steps are assumptions; break-even needs 1,824 resolved if hiring falls ROI.

Why should we care?

- Next year, 51,750 contacts would need about 138 agents, not 120 GUIDE D4
- Fraud waits in the same queue as repeat questions PRD
- Each resolved contact saves about 9.6 agent minutes B2
- Freeing about 5.8 agents' workload in the base case STATUS
- A leak breaks banking law, not just privacy REG

KPI tree

Lower cost per contact, capacity for 15% growth GUIDE

→ In-scope share D2 · Chat adoption C1 · Minutes saved B2 · Freed time used B3 →

Verified resolution, 40% PRD

Fraud recall, 100% PRD

Grounded, ≥90% PRD

p95 under 6 s STATUS

PROBLEM SPACE

How do people get answers today?

How people solve it today

- Call the contact center and wait in the queue GUIDE
- Chat with an agent, who searches policy and transaction screens GUIDE
- Walk into a branch MKT
- App and FAQ pages: generic, not about their own transaction GUIDE
- IVR "press 1" menus MKT
- The agent explains, quality varies, and notes are rarely analysed GUIDE

How competitors solve it

Who	What they do	Gap
Egyptian bank bots	Product and service questions; NBE asks users not to share personal data	No personal, grounded answers MKT
Intercom Fin	$0.99 per resolution; claims 67% resolution	Foreign SaaS; "resolved" counts people who leave MKT
Agentforce	$2 per conversation; acts in CRM	Needs Service Cloud; highest price MKT
Klarna	Resolution 11 min to under 2 min	Quality fell; rehired people MKT
Zendesk AI agents	About $1.50 to $2 per automated resolution	Cost grows with success MKT
Kore.ai, Intella	Arabic at Middle East banks; Arabic-dialect voice	Enterprise contracts; chat depth unproven MKT
Bank of America Erica	The global benchmark assistant	Took 8 years; customer version not generative MKT

Tools people use

Phone · Bank app · WhatsApp · Facebook page · Branch

Customers message the bank's Facebook page or walk in MKT. WhatsApp is planned for release R3, not built PLAY.

The brief says the app is used for balance checks and little else, so in-app chat is a risk C1 A2.

Target segment

Nova Bank customers

Retail, logged in to the app ✅ chosen PRD

SMB customers: phase 2 GUIDE

Phone-only callers: not yet A3

Why retail: the 3 questions are theirs, and they write Arabic, English or Franco PRD.

Triple Fit scores STATUS

4/5 market **4/5** solution **3/5** incentive

Main problems, as hypotheses

🤔 Most of these contacts need **information, not an action** D3 Not yet tested

🤔 Customers with these questions **will use chat** in the app C1 Not yet tested

🤔 The answer **exists** in their records and Nova's rules S8 synthetic pass

🤔 IT can give a **read-only, real-time link** to one customer's records S1 Not yet tested

🤔 The Central Bank allows data to **reach the AI provider** R1 Not yet tested

RESEARCH

What do we know, and what is still missing?

Surveys and interviews

0 **No customer survey or bank-employee interview has been run yet.** Not yet tested STATUS

Planned survey (3 to 5 customers) GUIDE: when did you last contact your bank, and why? Would you trust a chat assistant with your dispute status? Arabic, English or Franco? What makes you ask for a person at once?

Planned interview: "Of your last 20 calls, how many needed no action?" D3

27 open information gaps, 12 of them high priority, such as the real contact split and agent cost GUIDE

If only one thing: 500 to 2,000 real anonymised messages. They test the language, the intents and the fraud check at once C2 R2.

Shadow run on 200 tickets

The copilot read 200 past tickets and we checked what it would have done STATUS.

synthetic data

Run 1 — **23%** Stage 2 tickets

Run 2 — **59%** linked to records

Run 3 — **63%** + new facts (1.0.11)

0/16 fraud tickets missed STATUS

1,207 agent minutes saved, run 3 STATUS

63% is an **answer rate**, not verified resolution. 33 of the 50 approved examples appear word for word in these tickets STATUS M2.

📋 **Shadow run details**

What customers write

No real quotes yet Not yet tested. These are test phrasings we wrote sample C2:

- *"Why was I charged EGP 25 on 14/9?"*
- *"ليه اتخصم مني 25 جنيه يوم 14/9؟"*
- *"3ayez a3raf el 25 geneh dol eh"*
- *"my card got declined in Paris just now"*

Secondary research

+14% issues resolved per hour with AI help for agents; +34% for new agents (5,179 agents) MKT

67% resolution claimed by Intercom Fin, a vendor claim MKT

2 min Klarna's resolution time, from 11; quality later fell MKT

15–40% automation of high-volume tasks at a Middle East bank MKT

Key takeaways

- Speed gains are easy; resolution and satisfaction are not MKT
- Measure verified resolution, not deflection GUIDE
- A warm handoff with a summary stops angry repeats MKT
- The best ROI evidence is for helping agents, so start agents-first MKT STATUS
- Data kept in Egypt is a real difference MKT R1
- Our own numbers are hypotheses until tested on real data A12
- Next: 1 or 2 bank interviews, public app reviews, a classmate test STATUS

USERS

Who is it for, and what do they go through?

Primary persona: the Nova customer sample



Retail customer, logged in to the app. Writes in Egyptian Arabic, English or Franco-Arabic PRD.

"When I have a question about my own fee, dispute or card, give me the right answer now, in my language." PRD

Arabic RTL | Franco | wants a person for fraud

Pains: waits behind repeat questions; FAQs are generic; fears being stuck with a bot or data misuse PRD GUIDE.

Goals: a sourced answer now; a person at once when something is wrong PRD.

Secondary persona: the support agent sample



Works the handoff inbox. Fewer repetitive calls, good handoff summaries GUIDE.

Fears: job loss; fixing bad bot answers GUIDE O1.

Fraud, Complaints, Agent queues | Suggest-only mode | KPIs reset for harder calls O2

Pains: searches policy and transaction screens by hand; quality varies GUIDE.

Goals: the full picture on handoff, so the customer never repeats themselves GUIDE O3.

Journey map: today vs with the copilot

	Has a question	Identity check	Finding the answer	Answering	Needs an action	Learning
Activities (today)	Calls or chats, waits in queue	Agent asks security questions	Agent searches policy and transactions	Agent explains; quality varies	Agent acts or transfers	Notes in a ticket, rarely analysed
Emotions sample	😟	😐	😠	😕	😟	😐
Thoughts sample	"Why was I charged?"	"Again these questions?"	"How long is this?"	"Is that right?"	"Will someone fix it?"	"Will it happen again?"
Touchpoints	📞 📱 💬	📞	💻	📱	🧑‍💼	📋
Our solution	Chat any time; suggested questions UX	Already logged in PRD R6	Record picker, investigator 1.0.12 1.0.13	Cited answer, "Why this answer?" UX	Handoff with summary and reference PRD O3	Every turn logged, 👍/👎 UX

Activities and solution rows from the project guide and UX notes GUIDE UX. Emotions and thoughts are illustrative.

Job to be done

"When I have a question about my own fee, dispute or card, give me the right answer now, in my language, from the bank's real rules. When something is wrong, like a payment I didn't make, get me to a person immediately." PRD

Other people it touches

🧑‍💼 contact-center head | ⚖️ compliance and legal | 🕵️ risk and fraud | 💳 disputes and card ops | 💻 IT and security | 🪙 finance

Roles assumed, not interviewed GUIDE.

THE PROBLEM

What is the real problem?

The true problem: 5 whys

Problem statement: customers contact the bank for dispute status, fee explanations and card replacement because the app does not answer these questions: about 27,000 avoidable contacts a month, growing 15% a year GUIDE.

1. **Why?** Customers call because they cannot find the answer themselves.
2. **Why?** The app shows balances only, and FAQs are generic.
3. **Why?** The question is personal: "why was I charged 75 EGP on 4 September?"
4. **Why?** Agents answer by reading the same policy plus the customer's transactions, a pattern AI can follow.
5. **Why?** Hypothesis: most of these contacts need information, not a decision.

Step 5 is unproven D3 **Not yet tested**

It will never: move money, waive fees, decide disputes, give credit or financial advice, or answer for anyone but the logged-in customer PRD.

Is it a real problem?

- The brief: 60% of 45,000 contacts are 3 questions PRD D2, with no breakdown yet A5
- No Egyptian bank bot found answering "why was I charged this?" from the customer's own transactions MKT
- Disputes and lost cards often need an action or a person's reassurance A5 D3
- People evidence: none yet; bank interviews planned **Not yet tested**

Why solve it now?

- Contacts grow 15% a year, faster than headcount GUIDE D4
- The market is consolidating: Salesforce agreed to buy Fin MKT
- The window is real but short: Intella with Visa, Kore.ai in Middle East banks MKT
- Egypt's second AI strategy (2025 to 2030) and trustworthy-AI guidelines (Mar 2026) REG

Value generated

For users

- An answer any time, in their language PRD
- Every answer shows its source PRD UX
- A person one tap away, always UX
- Fraud goes straight to the Fraud team PRD

For the company

- About 5.8 of 120 agents' workload freed STATUS
- AI costs about EGP 0.024 an answer B5
- An audit trail for every answer GUIDE
- Break-even: 1,824 to 4,869 resolved a month ROI

+18% ROI if hiring really falls ROI

-56% on minutes saved only ROI

The value is real only if the bank hires fewer people or moves agents to other work. Idle freed time saves nothing B3.

SOLUTIONS

Which solution did we choose, and why?

How might we…

…give a logged-in customer the right answer about their own fee, dispute or card, without inventing anything, and without trapping them with a bot?

6 types reviewed MKT → **2** compared → **1** chosen

Chosen: grounded answers, with actions kept for people and a path to agent help MKT.

Four product decisions GUIDE: 3 use cases in, the rest out · a warm handoff with rules · Arabic and English from day one, Franco too · one metric, verified resolution.

Top 2 solutions compared

	Grounded answer bot (chosen)	Agent-assist copilot
First response	Seconds	Unchanged: a person answers
Resolution	Minutes for information questions	Shorter handling per contact
Cost	Per token; little scripting	Per agent seat
Risk	Risk if it guesses	Agent stays in control
Evidence	Vendor claims	Strongest: +14% an hour

From the six-type table MKT. The pilot starts agents-first STATUS.

Shortlisted initiatives

Trusted answers score 12 — Build now

Safe handoff score 15 — Build now

Real evidence score 29 — Build now

Document control score 9 — Park: next

Bank connection score 6 — Park: next

On-premise and actions score 4 — Park: later

Weighted scores and Now, Next, Later from the delivery playbook, a proposal PLAY. Built features: record picker, investigator, handoff summary, chat memory STATUS.

Why it might work

🔁 Current behaviour

- 📞 Customers already ask these 3 questions PRD
- 💻 Agents answer from the same policy plus transactions, a pattern AI can follow GUIDE
- 📱 The customer taps the exact record instead of typing details STATUS

🧠 User psychology

- 🛡️ A first-message card sets what it can and can't do UX
- 🔍 "Why this answer?" earns trust UX
- 🙋 Control: "Talk to a person" is always visible UX
- 👍 👍/👎 under every answer; 👎 offers a person UX

Lesson from Klarna: escalated customers arrived angry because they had to repeat themselves. A handoff summary is cheap and fixes this MKT.

Benefits for the company

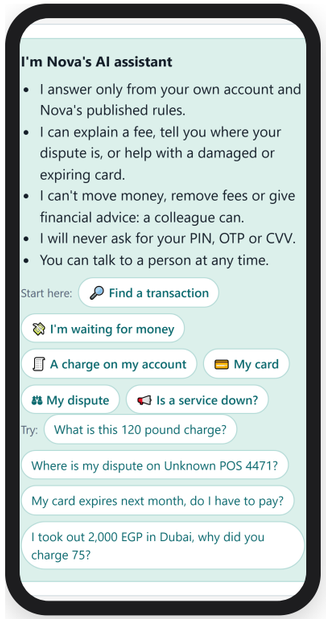
- 👥 Capacity for growth, not layoffs: about 18 hires avoided GUIDE
- ⚙️ One adapter: the model can be swapped; learning lives in Nova's files PRD
- 🔒 Model, prompt and sources stored with each answer GUIDE
- 🪙 AI is the smallest cost: about EGP 81 a month ROI
- 🛡️ Kill switch: one setting sends every message to a person PRD

HOW IT LOOKS · 1

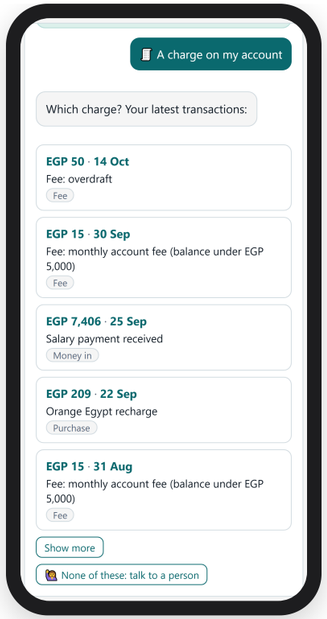
What does the customer see?

The customer chat at phone width, step by step

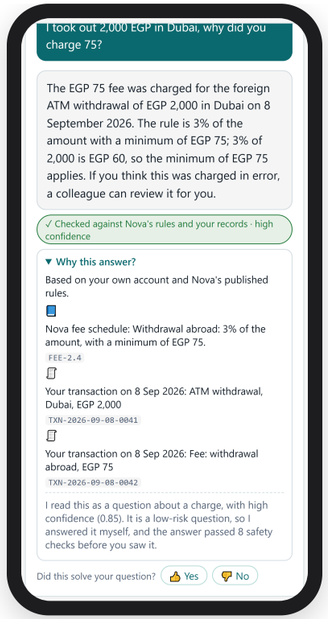
synthetic data



1 Welcome card: what it can and can't do



2 Tap the exact record, no typing



3 Cited answer; "Why this answer?" shows rule and records

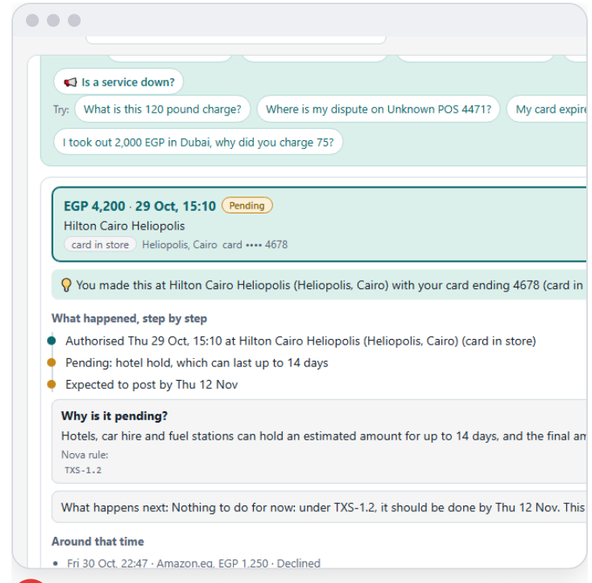


4 Franco in, Arabic out; fraud to a person



5 No source? It won't guess; a person, with a reference

Transaction investigator



6 One transaction, step by step: why it is pending and what happens next

1.0.13 STATUS Also: "I'm waiting for money" shows where incoming money is STATUS.

Key screens

Fee answer
Names the fee and quotes the schedule line, e.g. FEE-2.4 (screen 3) PRD

Dispute status
Stage and decision target date: 45 days, 90 for card-network cases STATUS

Card
Replacement process and fee; "Block card" asks to confirm and has Cancel UX

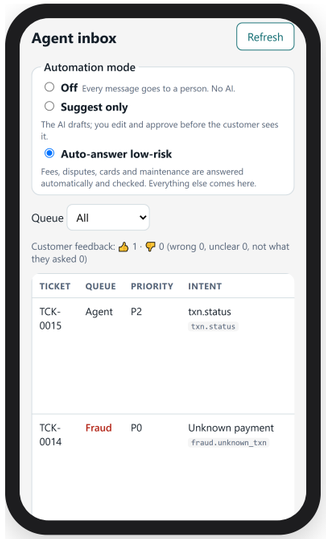
Waiting for money
Where incoming money is, and when to expect it STATUS

Try the prototype (runs locally) · Demo script · Stage 6: integration and UX · UX principles · Investigator spec

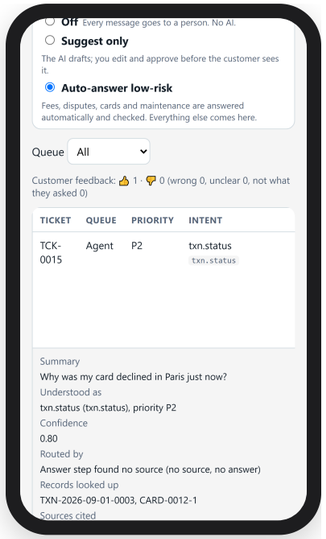
HOW IT LOOKS · 2

What do agents and the PM see?

Behind the handoff (synthetic data)

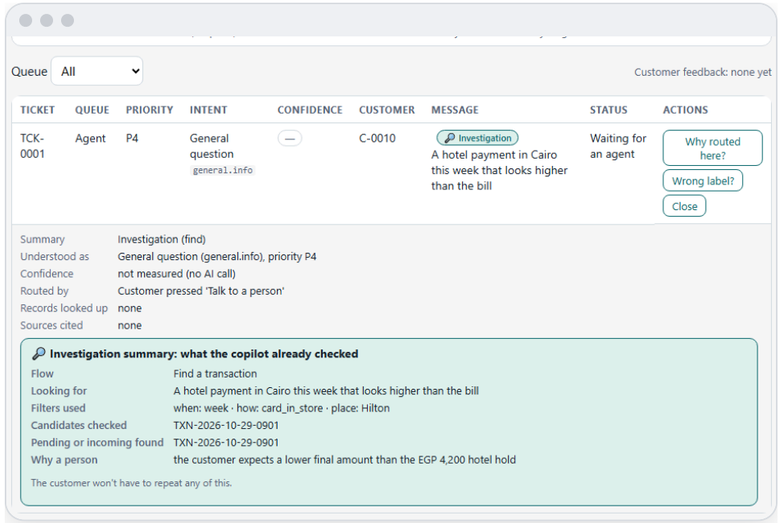


1 Inbox: modes, queues, priority



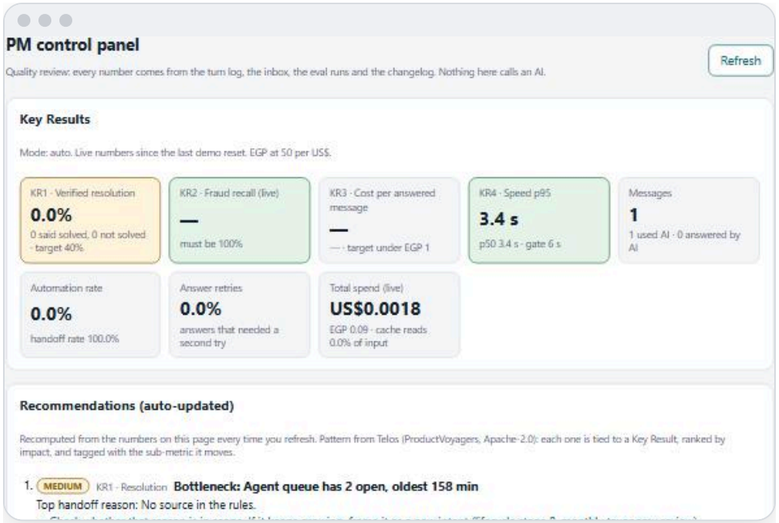
2 "Why routed here?": no source, no answer

Investigation summary for the agent



4 What the copilot already checked, so the customer never repeats

PM control panel



5 Key results, live recommendations, cost by model

Three automation modes

- 🛑 **Off:** every message to a person, no AI UX
- 🤵 **Suggest only:** the AI drafts, an agent approves. The pilot starts here UX STATUS
- 🤖 **Auto-answer low-risk:** checked answers go straight out UX
- 📝 Every mode change is logged UX

What the agent gets on handoff

- 📋 Intent, priority, confidence, records looked up and sources cited UX
- 🗓️ A ticket reference for every handoff and complaint PRD
- 📌 One honest line on why: not sure, needs a person, for your safety… UX
- 👍 Label corrections are saved; nothing changes until a person approves UX

User stories and docs

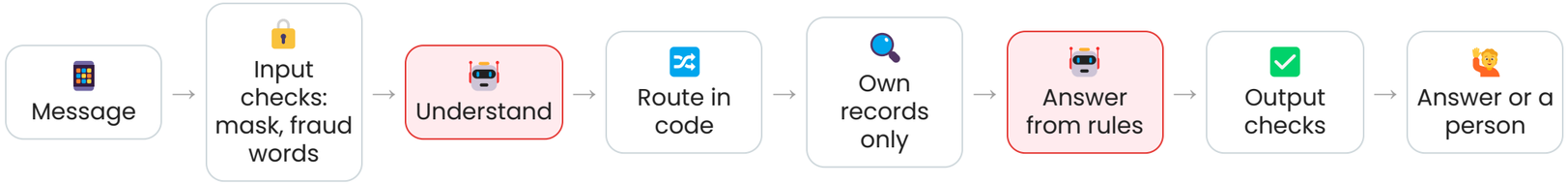
The delivery playbook walks 6 features through 8 team views, with stories and acceptance criteria PLAY.

📋 User stories (playbook) · 📁 Document control · 🛠️ How we built it · 📊 Stage 8: monitoring

OPERATIONS AND SUCCESS

How does it run, and how will we know it worked?

How the MVP runs behind the scenes



One message, 7 steps; only the 2 red steps use AI (Claude Haiku 5.5) PRD STATUS. The model is never the security boundary PRD.

Success metrics

Level	Metric	Target	Today	Why
North Star	Verified resolution rate	**40%**	**Not yet tested**	Counts customers who don't come back in 7 days
L1	Fraud recall	100%	13/13 STATUS	Fraud must never wait
L1	Grounded answers	≥ 90%	97.1% STATUS	No source, no answer
L2	Intent accuracy	≥ 90%	96.9% STATUS	Right route, right rule
Adoption	In-scope contacts that start in chat	20%	**Not yet tested** C1	Drives every saving
Engagement	"Did this solve your question?" answered	Enough to measure	**Not yet tested** C6	Avoids a happy-only bias
Guardrail	Security, harmless	All pass	18/18, 7/7 STATUS	Zero leaks, zero unsafe actions
Operations	p95 reply time	< 6 s	5.5 s STATUS	Slow replies push people to call

What can go wrong, and the fix

Risk	Mitigation
Another customer's data shown R6	Customer ID enforced in the lookup; security 18/18
A fraud report missed R2	Fraud words in code, no AI call; 13/13
An invented fee or promise PRD	Output checks; grounded 97.1%
Data cannot leave Egypt R1	Compliance check; on-premise phase
Agents resist it O1	Agents first: Suggest-only pilot
The 0.6 rule is wrong STATUS	Logged as near_sure_answer; agent approves

Stop or redesign if, after 8 weeks of pilot, verified resolution is under 15%, CSAT drops more than 10% below the human channel, or any data leak is confirmed GUIDE.

Thank you Next: evidence from real people, then an agent-first pilot STATUS.

Project map | Prototype | All slides | Stage reports